

Lovely Professional University, Punjab

Course Code	Course Title	Lectures	Tutorials	Practicals	Credits	
CSE472	DEEP LEARNING FOR NATURAL LANGUAGE PROCESSING	2	0	2	3	
Course Weightage	ATT: 5 CA: 45 ETT: 50	Exam Category: X1: Mid Term Exam: Not Applicable – End Term Exam: All MCQ				

Course Outcomes :Through this course students should be able to

CO1 :: Understand the origin, evolution, and linguistic foundations of Natural Language Processing

CO2 :: Understand word embeddings, semantic relationships, and embedding visualization techniques

CO3 :: Apply deep learning sequence models for NLP to text and sentiment classification tasks

CO4 :: Apply seq2seq models with attention for translation and summarization tasks

CO5 :: Use transformer architectures and pretrained language models for fine-tuned NLP tasks

CO6 :: Apply generative NLP and large language models for text generation and evaluation tasks, considering explainability and hallucination

	TextBooks (T)		
Sr No	Title	Author	Publisher Name
T-1	GETTING STARTED WITH DEEP LEARNING FOR NATURAL LANGUAGE PROCESSING	SUNIL PATEL	BPB Publication, New Delhi

	Reference Books (R)		
Sr No	Title	Author	Publisher Name
R-1	SPEECH AND LANGUAGE PROCESSING: AN INTRODUCTION TO NATURAL LANGUAGE PROCESSING, COMPUTATIONAL LINGUISTICS AND SPEECH RECOGNITION	DANIEL JURAFSKY	Pearson Education India

Other Reading (OR)	
Sr No	Journals articles as Compulsary reading (specific articles, complete reference)
OR-1	Daniel Jurafsky & James H. Martin, Speech and Language Processing (3rd Edition Draft). ,
OR-2	Steven Bird, Ewan Klein, & Edward Loper, Natural Language Processing with Python (O'Reilly). ,

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OR-3	Jacob Eisenstein, Introduction to Natural Language Processing (MIT Press). ,
OR-4	Delip Rao & Brian McMahan, Natural Language Processing with PyTorch. ,
OR-5	Yoav Goldberg, Neural Network Methods for Natural Language Processing. ,

Relevant Websites (RW)

Sr No	(Web address) (only if relevant to the course)	Salient Features
RW-1	Stanford NLP Group – https://nlp.stanford.edu/	Cutting-edge NLP Research
RW-2	Hugging Face – https://huggingface.co/	Pretrained Models Repository
RW-3	NLTK Documentation – https://www.nltk.org/	Comprehensive NLP Toolkit
RW-4	spaCy Documentation – https://spacy.io/	Industrial-Strength NLP Library
RW-5	TensorFlow Text – https://www.tensorflow.org/text	Text Processing Extensions
RW-6	PyTorch – https://pytorch.org/	Flexible Deep Learning
RW-7	Kaggle NLP Datasets – https://www.kaggle.com/datasets	Large NLP Datasets
RW-8	Google AI – https://ai.google/	Advanced AI Research
RW-9	Papers with Code (NLP) – https://paperswithcode.com/	Benchmarks with Source Code

Audio Visual Aids (AV)

Sr No	(AV aids) (only if relevant to the course)	Salient Features
AV-1	Stanford CS224N: Natural Language Processing with Deep Learning (YouTube)	NLP with deep learning
AV-2	DeepLearning.AI – NLP Specialization (Coursera)	Deep Learning
AV-3	Hugging Face YouTube Channel	Pretrained Models Repository
AV-4	freeCodeCamp NLP Tutorials	NLP
AV-5	MIT OpenCourseWare – NLP lectures	NLP
AV-6	Google Developers AI/ML videos	AI ML

Software/Equipments/Databases

Sr No	(S/E/D) (only if relevant to the course)	Salient Features
SW-1	Python	versatile programming language
SW-2	Jupyter Notebook	Interactive coding environment
SW-3	Google Colab	Cloud-based notebook platform
SW-4	NLTK	NLP toolkit
SW-5	spaCy	Fast industrial NLP

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SW-6	Gensim	Topic modeling library
SW-7	Hugging Face Transformers	Pre-trained transformer models
SW-8	TensorFlow	Deep learning framework
SW-9	PyTorch	Dynamic neural networks
SW-10	Scikit-learn	Machine learning library
SW-11	OpenAI API	Advanced language models
SW-12	LangChain	LLM application framework
SW-13	VS Code	code editor
SW-14	PyCharm	Python IDE support
SW-15	Git & GitHub	Version control collaboration

Virtual Labs (VL)		
Sr No	(VL) (only if relevant to the course)	Salient Features
VL-1	Google Colab (Cloud-based NLP experimentation)	NLP
VL-2	Kaggle Notebooks	Kaggle
VL-3	Hugging Face Spaces	Pre-trained transformer models
VL-4	TensorFlow Playground (for neural network concepts)	TensorFlow
VL-5	IBM Skills Network Labs	Hands-on cloud labs
VL-6	Azure AI Language Studio	NLP development
VL-7	Google Vertex AI Workbench	ML notebooks
VL-8	AWS SageMaker Studio Lab	ML environment

LTP week distribution: (LTP Weeks)	
Weeks before MTE	7
Weeks After MTE	7
Spill Over (Lecture)	4

Detailed Plan For Lectures

Week Number	Lecture Number	Broad Topic(Sub Topic)	Chapters/Sections of Text/reference books	Other Readings, Relevant Websites, Audio Visual Aids, software and Virtual Labs	Lecture Description	Learning Outcomes	Pedagogical Tool Demonstration/ Case Study / Images / animation / ppt etc. Planned	Live Examples

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Week 1	Lecture 1	Foundations of NLP(origin of NLP)	T-1 R-1		This lecture introduces Natural Language Processing (NLP) by discussing its origin and historical evolution, along with an overview of major real-world applications of NLP.	Students will be able to explain the origin and evolution of NLP and identify major application areas where NLP is used.	Interactive lecture, timeline-based explanation, discussion using real-world NLP systems.	Google Search, Google Translate, chatbots, voice assistants such as Alexa and Siri.
		Foundations of NLP (applications of NLP)	T-1 R-1	OR-1 RW-1	This lecture introduces Natural Language Processing (NLP) by discussing its origin and historical evolution, along with an overview of major real-world applications of NLP.	Students will be able to explain the origin and evolution of NLP and identify major application areas where NLP is used.	Interactive lecture, timeline-based explanation, discussion using real-world NLP systems.	Google Search, Google Translate, chatbots, voice assistants such as Alexa and Siri.
	Lecture 2	Foundations of NLP (language and grammar)	T-1 R-1		This lecture focuses on the fundamentals of language and grammar, covering linguistic essentials required for understanding natural language in NLP systems.	Students will be able to understand language structure and explain fundamental linguistic concepts essential for Natural Language Processing.	Concept explanation with examples, sentence analysis, interactive discussion.	Grammatical sentence structures, ambiguity in natural language sentences.
		Foundations of NLP (linguistic essentials)	T-1 R-1	OR-2 RW-2	This lecture focuses on the fundamentals of language and grammar, covering linguistic essentials required for understanding natural language in NLP systems.	Students will be able to understand language structure and explain fundamental linguistic concepts essential for Natural Language Processing.	Concept explanation with examples, sentence analysis, interactive discussion.	Grammatical sentence structures, ambiguity in natural language sentences.
Week 2	Lecture 3	Foundations of NLP (morphology)	R-1		This lecture covers the linguistic concepts of morphology and syntax, explaining how words are formed and how sentence structures are represented for Natural Language Processing.	Students will be able to understand morphological and syntactic structures of language and explain their importance in Natural Language Processing tasks.	Concept explanation using examples, sentence structure analysis, guided discussion.	Word formation examples, sentence parsing trees, syntactic ambiguity in sentences.

Week 2	Lecture 3	Foundations of NLP(syntax)	R-1	OR-3 RW-3	This lecture covers the linguistic concepts of morphology and syntax, explaining how words are formed and how sentence structures are represented for Natural Language Processing.	Students will be able to understand morphological and syntactic structures of language and explain their importance in Natural Language Processing tasks.	Concept explanation using examples, sentence structure analysis, guided discussion.	Word formation examples, sentence parsing trees, syntactic ambiguity in sentences.
	Lecture 4	Foundations of NLP (semantics)	T-1		This lecture focuses on semantics in natural language and discusses the major challenges faced by Natural Language Processing systems, including ambiguity, context understanding, and language variability.	Students will be able to explain semantic aspects of language and identify key challenges involved in designing effective NLP systems.	Concept explanation with examples, discussion on ambiguity and context, problem-based learning.	Word sense ambiguity, sarcasm in text, context-dependent meaning in sentences.
		Foundations of NLP (challenges of NLP)	T-1		This lecture focuses on semantics in natural language and discusses the major challenges faced by Natural Language Processing systems, including ambiguity, context understanding, and language variability.	Students will be able to explain semantic aspects of language and identify key challenges involved in designing effective NLP systems.	Concept explanation with examples, discussion on ambiguity and context, problem-based learning.	Word sense ambiguity, sarcasm in text, context-dependent meaning in sentences.
Week 3	Lecture 5	Text Processing(Bag-of-Words)	T-1		This lecture introduces classical vector space representations of text using Bag-of-Words, n-gram models, and TF-IDF for representing textual data numerically.	Students will be able to represent text using Bag-of-Words, n-gram, and TF-IDF models and explain their role as vector space representations in NLP.	Concept explanation with numerical examples, board-based derivation, and worked examples.	Creating document-term matrices and TF-IDF vectors for sample text documents.
		Text Processing(n-grams)	T-1	OR-4 RW-4 RW-5	This lecture introduces classical vector space representations of text using Bag-of-Words, n-gram models, and TF-IDF for representing textual data numerically.	Students will be able to represent text using Bag-of-Words, n-gram, and TF-IDF models and explain their role as vector space representations in NLP.	Concept explanation with numerical examples, board-based derivation, and worked examples.	Creating document-term matrices and TF-IDF vectors for sample text documents.

Week 3	Lecture 5	Text Processing(TF-IDF)	T-1		This lecture introduces classical vector space representations of text using Bag-of-Words, n-gram models, and TF-IDF for representing textual data numerically.	Students will be able to represent text using Bag-of-Words, n-gram, and TF-IDF models and explain their role as vector space representations in NLP.	Concept explanation with numerical examples, board-based derivation, and worked examples.	Creating document-term matrices and TF-IDF vectors for sample text documents.
	Lecture 6	Text Processing (tokenization)	R-1	OR-5 RW-6	This lecture introduces fundamental text preprocessing techniques including tokenization, normalization, stop-word removal, and punctuation handling, which are essential for preparing raw text data for NLP tasks.	Students will be able to preprocess raw textual data using tokenization, normalization, stop-word removal, and punctuation handling techniques.	Demonstration-based teaching, step-by-step preprocessing walkthrough, hands-on examples.	Preprocessing a paragraph of text using Python libraries such as NLTK or spaCy.
		Text Processing(stop-word removal)	R-1		This lecture introduces fundamental text preprocessing techniques including tokenization, normalization, stop-word removal, and punctuation handling, which are essential for preparing raw text data for NLP tasks.	Students will be able to preprocess raw textual data using tokenization, normalization, stop-word removal, and punctuation handling techniques.	Demonstration-based teaching, step-by-step preprocessing walkthrough, hands-on examples.	Preprocessing a paragraph of text using Python libraries such as NLTK or spaCy.
		Text Processing(punctuation handling)	R-1		This lecture introduces fundamental text preprocessing techniques including tokenization, normalization, stop-word removal, and punctuation handling, which are essential for preparing raw text data for NLP tasks.	Students will be able to preprocess raw textual data using tokenization, normalization, stop-word removal, and punctuation handling techniques.	Demonstration-based teaching, step-by-step preprocessing walkthrough, hands-on examples.	Preprocessing a paragraph of text using Python libraries such as NLTK or spaCy.

Week 3	Lecture 6	Text Processing (normalization)	R-1		This lecture introduces fundamental text preprocessing techniques including tokenization, normalization, stop-word removal, and punctuation handling, which are essential for preparing raw text data for NLP tasks.	Students will be able to preprocess raw textual data using tokenization, normalization, stop-word removal, and punctuation handling techniques.	Demonstration-based teaching, step-by-step preprocessing walkthrough, hands-on examples.	Preprocessing a paragraph of text using Python libraries such as NLTK or spaCy.
Week 4	Lecture 7	Text Processing(stemming)	R-1	RW-7 RW-8 RW-9	This lecture discusses advanced text preprocessing techniques including stemming, lemmatization, and handling out-of-vocabulary words to improve the quality of text representation in NLP systems.	Students will be able to apply stemming and lemmatization techniques and handle out-of-vocabulary words during text preprocessing for NLP tasks.	Concept explanation with comparative examples, demonstration using NLP libraries, guided practice.	Comparison of stemming vs lemmatization results and handling unknown words in real text datasets.
		Text Processing (lemmatization)	R-1		This lecture discusses advanced text preprocessing techniques including stemming, lemmatization, and handling out-of-vocabulary words to improve the quality of text representation in NLP systems.	Students will be able to apply stemming and lemmatization techniques and handle out-of-vocabulary words during text preprocessing for NLP tasks.	Concept explanation with comparative examples, demonstration using NLP libraries, guided practice.	Comparison of stemming vs lemmatization results and handling unknown words in real text datasets.
		Text Processing(handling out-of-vocabulary words)	R-1		This lecture discusses advanced text preprocessing techniques including stemming, lemmatization, and handling out-of-vocabulary words to improve the quality of text representation in NLP systems.	Students will be able to apply stemming and lemmatization techniques and handle out-of-vocabulary words during text preprocessing for NLP tasks.	Concept explanation with comparative examples, demonstration using NLP libraries, guided practice.	Comparison of stemming vs lemmatization results and handling unknown words in real text datasets.

Week 4	Lecture 8	Word Embeddings and Vector Representations (vector space models)	T-1 R-1		This lecture introduces vector space models and dense word embeddings, focusing on the Word2Vec architecture and its CBOW and Skip-Gram models for learning distributed word representations.	Students will be able to explain vector space representations and describe how Word2Vec models such as CBOW and Skip-Gram learn dense word embeddings.	Conceptual explanation using diagrams, walkthrough of Word2Vec architecture, and illustrative examples.	Training a Word2Vec model on a small corpus and observing similarity between word vectors.
		Word Embeddings and Vector Representations (dense word embeddings)	T-1 R-1		This lecture introduces vector space models and dense word embeddings, focusing on the Word2Vec architecture and its CBOW and Skip-Gram models for learning distributed word representations.	Students will be able to explain vector space representations and describe how Word2Vec models such as CBOW and Skip-Gram learn dense word embeddings.	Conceptual explanation using diagrams, walkthrough of Word2Vec architecture, and illustrative examples.	Training a Word2Vec model on a small corpus and observing similarity between word vectors.
		Word Embeddings and Vector Representations (Word2Vec)	T-1 R-1		This lecture introduces vector space models and dense word embeddings, focusing on the Word2Vec architecture and its CBOW and Skip-Gram models for learning distributed word representations.	Students will be able to explain vector space representations and describe how Word2Vec models such as CBOW and Skip-Gram learn dense word embeddings.	Conceptual explanation using diagrams, walkthrough of Word2Vec architecture, and illustrative examples.	Training a Word2Vec model on a small corpus and observing similarity between word vectors.
		Word Embeddings and Vector Representations (CBOW)	T-1 R-1		This lecture introduces vector space models and dense word embeddings, focusing on the Word2Vec architecture and its CBOW and Skip-Gram models for learning distributed word representations.	Students will be able to explain vector space representations and describe how Word2Vec models such as CBOW and Skip-Gram learn dense word embeddings.	Conceptual explanation using diagrams, walkthrough of Word2Vec architecture, and illustrative examples.	Training a Word2Vec model on a small corpus and observing similarity between word vectors.

Week 4	Lecture 8	Word Embeddings and Vector Representations (Skip-Gram models)	T-1 R-1		This lecture introduces vector space models and dense word embeddings, focusing on the Word2Vec architecture and its CBOW and Skip-Gram models for learning distributed word representations.	Students will be able to explain vector space representations and describe how Word2Vec models such as CBOW and Skip-Gram learn dense word embeddings.	Conceptual explanation using diagrams, walkthrough of Word2Vec architecture, and illustrative examples.	Training a Word2Vec model on a small corpus and observing similarity between word vectors.
Week 5	Lecture 9	Word Embeddings and Vector Representations (GloVe embeddings)	T-1		This lecture explains GloVe word embeddings and demonstrates how semantic similarity and analogy relationships are captured using distributed word representations.	Students will be able to explain GloVe embeddings and analyze semantic similarity and analogy relationships using word embeddings.	Concept explanation with mathematical intuition, example-based learning, discussion.	Word analogy tasks such as king - man + woman ~ queen and similarity comparison between word pairs.
		Word Embeddings and Vector Representations (capturing semantic similarity)	T-1		This lecture explains GloVe word embeddings and demonstrates how semantic similarity and analogy relationships are captured using distributed word representations.	Students will be able to explain GloVe embeddings and analyze semantic similarity and analogy relationships using word embeddings.	Concept explanation with mathematical intuition, example-based learning, discussion.	Word analogy tasks such as king - man + woman ~ queen and similarity comparison between word pairs.
		Word Embeddings and Vector Representations (analogy relationships)	T-1		This lecture explains GloVe word embeddings and demonstrates how semantic similarity and analogy relationships are captured using word vectors. It also introduces visualization of high-dimensional embedding spaces using dimensionality reduction techniques such as PCA and t-SNE.	Students will be able to explain GloVe embeddings, analyze semantic similarity and analogy relationships, and visualize word embedding spaces using PCA and t-SNE techniques.	Concept explanation with examples, mathematical intuition for embeddings, visualization-based demonstration, and guided discussion.	Word analogy tasks (king - man + woman ~ queen) and PCA/t-SNE plots showing clustering of semantically similar words.

Week 5	Lecture 9	Word Embeddings and Vector Representations (visualizing embedding spaces using PCA or t-SNE)	T-1	AV-4 AV-5 AV-6	This lecture explains GloVe word embeddings and demonstrates how semantic similarity and analogy relationships are captured using word vectors. It also introduces visualization of high-dimensional embedding spaces using dimensionality reduction techniques such as PCA and t-SNE.	Students will be able to explain GloVe embeddings, analyze semantic similarity and analogy relationships, and visualize word embedding spaces using PCA and t-SNE techniques.	Concept explanation with examples, mathematical intuition for embeddings, visualization-based demonstration, and guided discussion.	Word analogy tasks (king - man + woman ~ queen) and PCA/t-SNE plots showing clustering of semantically similar words.
	Lecture 10				Test - Code based			
Week 6	Lecture 11	Deep Learning Sequence Models for NLP(sequential text data)	T-1	AV-1 AV-2 AV-3	This lecture introduces sequential text data and recurrent neural networks, explaining how sequence information is modeled in Natural Language Processing tasks.	Students will be able to understand sequential text data and explain the working principles of recurrent neural networks for NLP applications.	Conceptual explanation using sequence diagrams, step-by-step walkthrough of RNN architecture, discussion.	Sequence prediction examples and simple RNN-based text processing demonstrations.
		Deep Learning Sequence Models for NLP(recurrent neural networks)	T-1		This lecture introduces sequential text data and recurrent neural networks, explaining how sequence information is modeled in Natural Language Processing tasks.	Students will be able to understand sequential text data and explain the working principles of recurrent neural networks for NLP applications.	Conceptual explanation using sequence diagrams, step-by-step walkthrough of RNN architecture, discussion.	Sequence prediction examples and simple RNN-based text processing demonstrations.
	Lecture 12	Deep Learning Sequence Models for NLP(long short term memory networks)	T-1		This lecture explains Long Short-Term Memory (LSTM) networks and Gated Recurrent Units (GRUs), highlighting how they address the limitations of basic RNNs in modeling long-term dependencies.	Students will be able to explain the architecture and working of LSTM and GRU networks and compare them with basic RNN models for NLP tasks.	Diagram-based explanation, step-by-step cell analysis, comparative discussion between LSTM and GRU.	Sentiment classification example using LSTM and comparison with GRU-based models.

Week 6	Lecture 12	Deep Learning Sequence Models for NLP(gated recurrent units)	T-1		This lecture explains Long Short-Term Memory (LSTM) networks and Gated Recurrent Units (GRUs), highlighting how they address the limitations of basic RNNs in modeling long-term dependencies.	Students will be able to explain the architecture and working of LSTM and GRU networks and compare them with basic RNN models for NLP tasks.	Diagram-based explanation, step-by-step cell analysis, comparative discussion between LSTM and GRU.	Sentiment classification example using LSTM and comparison with GRU-based models.
Week 7	Lecture 13	Deep Learning Sequence Models for NLP (bidirectional RNNs)	T-1		This lecture discusses sequence training techniques including teacher forcing and truncated backpropagation through time, along with evaluation metrics used for assessing sequence models in NLP tasks.	Students will be able to apply sequence training techniques and evaluate deep learning sequence models using appropriate metrics.	Concept explanation with training diagrams, numerical examples, discussion-based learning.	Training loss curves, evaluation of sequence models using accuracy and perplexity.
		Deep Learning Sequence Models for NLP(sequence modeling applications)	T-1		This lecture discusses sequence training techniques including teacher forcing and truncated backpropagation through time, along with evaluation metrics used for assessing sequence models in NLP tasks.	Students will be able to apply sequence training techniques and evaluate deep learning sequence models using appropriate metrics.	Concept explanation with training diagrams, numerical examples, discussion-based learning.	Training loss curves, evaluation of sequence models using accuracy and perplexity.
		Deep Learning Sequence Models for NLP(sentiment classification)	T-1		This lecture discusses sequence training techniques including teacher forcing and truncated backpropagation through time, along with evaluation metrics used for assessing sequence models in NLP tasks.	Students will be able to apply sequence training techniques and evaluate deep learning sequence models using appropriate metrics.	Concept explanation with training diagrams, numerical examples, discussion-based learning.	Training loss curves, evaluation of sequence models using accuracy and perplexity.

Week 7	Lecture 13	Deep Learning Sequence Models for NLP(text classification)	T-1		This lecture discusses sequence training techniques including teacher forcing and truncated backpropagation through time, along with evaluation metrics used for assessing sequence models in NLP tasks.	Students will be able to apply sequence training techniques and evaluate deep learning sequence models using appropriate metrics.	Concept explanation with training diagrams, numerical examples, discussion-based learning.	Training loss curves, evaluation of sequence models using accuracy and perplexity.
		Deep Learning Sequence Models for NLP(sequence training techniques)	T-1		This lecture discusses sequence training techniques including teacher forcing and truncated backpropagation through time, along with evaluation metrics used for assessing sequence models in NLP tasks.	Students will be able to apply sequence training techniques and evaluate deep learning sequence models using appropriate metrics.	Concept explanation with training diagrams, numerical examples, discussion-based learning.	Training loss curves, evaluation of sequence models using accuracy and perplexity.
		Deep Learning Sequence Models for NLP(teacher forcing)	T-1		This lecture discusses sequence training techniques including teacher forcing and truncated backpropagation through time, along with evaluation metrics used for assessing sequence models in NLP tasks.	Students will be able to apply sequence training techniques and evaluate deep learning sequence models using appropriate metrics.	Concept explanation with training diagrams, numerical examples, discussion-based learning.	Training loss curves, evaluation of sequence models using accuracy and perplexity.
		Deep Learning Sequence Models for NLP(truncated backpropagation through time)	T-1		This lecture discusses sequence training techniques including teacher forcing and truncated backpropagation through time, along with evaluation metrics used for assessing sequence models in NLP tasks.	Students will be able to apply sequence training techniques and evaluate deep learning sequence models using appropriate metrics.	Concept explanation with training diagrams, numerical examples, discussion-based learning.	Training loss curves, evaluation of sequence models using accuracy and perplexity.

Week 7	Lecture 13	Deep Learning Sequence Models for NLP(evaluation metrics for sequence tasks)	T-1		This lecture discusses sequence training techniques including teacher forcing and truncated backpropagation through time, along with evaluation metrics used for assessing sequence models in NLP tasks.	Students will be able to apply sequence training techniques and evaluate deep learning sequence models using appropriate metrics.	Concept explanation with training diagrams, numerical examples, discussion-based learning.	Training loss curves, evaluation of sequence models using accuracy and perplexity.
		SPILL OVER						
Week 7	Lecture 14				Spill Over			
		MID-TERM						
Week 8	Lecture 15	Sequence-to-Sequence Models and Attention Mechanisms (encoder–decoder architectures for NLP)	T-1		This lecture introduces encoder–decoder architectures and sequence-to-sequence models, explaining their use in machine translation and text summarization tasks in Natural Language Processing.	Students will be able to explain encoder–decoder architectures and describe how sequence-to-sequence models are applied to machine translation and summarization tasks.	Architecture-based explanation using block diagrams, step-by-step flow discussion, interactive questioning.	Illustration of a basic sequence to-sequence model for simple translation and summarization tasks.
		Sequence-to-Sequence Models and Attention Mechanisms(sequence-to-sequence models for machine translation and summarization)	T-1		This lecture introduces encoder–decoder architectures and sequence-to-sequence models, explaining their use in machine translation and text summarization tasks in Natural Language Processing.	Students will be able to explain encoder–decoder architectures and describe how sequence-to-sequence models are applied to machine translation and summarization tasks.	Architecture-based explanation using block diagrams, step-by-step flow discussion, interactive questioning.	Illustration of a basic sequence to-sequence model for simple translation and summarization tasks.
	Lecture 16	Sequence-to-Sequence Models and Attention Mechanisms(attention in deep NLP)	T-1 R-1		This lecture introduces attention mechanisms in deep NLP and explains the concept of soft attention for improving sequence-to-sequence model performance.	Students will be able to explain the need for attention mechanisms and describe how soft attention improves sequence-to-sequence models in NLP.	Conceptual explanation with alignment diagrams, step-by step attention weight calculation, discussion.	Visualization of attention weights in a machine translation model.

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Week 8	Lecture 16	Sequence-to-Sequence Models and Attention Mechanisms(soft attention)	T-1 R-1		This lecture introduces attention mechanisms in deep NLP and explains the concept of soft attention for improving sequence-to-sequence model performance.	Students will be able to explain the need for attention mechanisms and describe how soft attention improves sequence-to-sequence models in NLP.	Conceptual explanation with alignment diagrams, step-by-step attention weight calculation, discussion.	Visualization of attention weights in a machine translation model.
Week 9	Lecture 17	Sequence-to-Sequence Models and Attention Mechanisms(alignment mechanisms)	T-1		This lecture explains Bahdanau and Luong attention mechanisms and demonstrates how attention is integrated into encoder–decoder networks to improve sequence-to-sequence performance	Students will be able to differentiate between Bahdanau and Luong attention mechanisms and explain how attention is integrated into encoder–decoder models.	Comparative explanation using diagrams, attention alignment visualization, discussion-based learning.	Comparison of Bahdanau and Luong attention using alignment visualizations in translation models.
		Sequence-to-Sequence Models and Attention Mechanisms(Bahdanau and Luong attention)	T-1	SW-1 SW-2 SW-3	This lecture explains Bahdanau and Luong attention mechanisms and demonstrates how attention is integrated into encoder–decoder networks to improve sequence-to-sequence performance	Students will be able to differentiate between Bahdanau and Luong attention mechanisms and explain how attention is integrated into encoder–decoder models.	Comparative explanation using diagrams, attention alignment visualization, discussion-based learning.	
		Sequence-to-Sequence Models and Attention Mechanisms(integrating attention into encoder–decoder networks)	T-1		This lecture explains Bahdanau and Luong attention mechanisms and demonstrates how attention is integrated into encoder–decoder networks to improve sequence-to-sequence performance	Students will be able to differentiate between Bahdanau and Luong attention mechanisms and explain how attention is integrated into encoder–decoder models.	Comparative explanation using diagrams, attention alignment visualization, discussion-based learning.	Comparison of Bahdanau and Luong attention using alignment visualizations in translation models.
	Lecture 18	Sequence-to-Sequence Models and Attention Mechanisms(evaluation techniques)	T-1 R-1		This lecture covers evaluation techniques for sequence-to-sequence models, including BLEU and ROUGE scores, and discusses the limitations of classical sequence-to-sequence architectures.	Students will be able to evaluate sequence-to-sequence models using BLEU and ROUGE metrics and identify the limitations of classical seq2seq models.	Metric-based explanation, worked examples, comparative discussion on model limitations.	BLEU and ROUGE score calculation for machine translation and summarization outputs.

Week 9	Lecture 18	Sequence-to-Sequence Models and Attention Mechanisms(BLEU and ROUGE scores)	T-1 R-1	VL-3 VL-4 VL-5	This lecture covers evaluation techniques for sequence-to-sequence models, including BLEU and ROUGE scores, and discusses the limitations of classical sequence-to-sequence architectures.	Students will be able to evaluate sequence to-sequence models using BLEU and ROUGE metrics and identify the limitations of classical seq2seq models.	Metric-based explanation, worked examples, comparative discussion on model limitations.	
		Sequence-to-Sequence Models and Attention Mechanisms(limitations of classical seq2seq models)	T-1 R-1		This lecture covers evaluation techniques for sequence-to-sequence models, including BLEU and ROUGE scores, and discusses the limitations of classical sequence-to-sequence architectures.	Students will be able to evaluate sequence to-sequence models using BLEU and ROUGE metrics and identify the limitations of classical seq2seq models.	Metric-based explanation, worked examples, comparative discussion on model limitations.	BLEU and ROUGE score calculation for machine translation and summarization outputs.
Week 10	Lecture 19	Transformers(transformer architecture)	T-1		This lecture introduces the Transformer architecture, focusing on self-attention and multi head attention mechanisms that enable parallel sequence modeling in NLP.	Students will be able to explain the Transformer architecture and describe how self attention and multi head attention mechanisms work in NLP models	Architecture-level explanation using block diagrams, attention visualization, concept-based discussion	Illustration of self-attention weight computation and comparison with recurrent models.
		Transformers(self-attention)	T-1		This lecture introduces the Transformer architecture, focusing on self-attention and multi head attention mechanisms that enable parallel sequence modeling in NLP.	Students will be able to explain the Transformer architecture and describe how self attention and multi head attention mechanisms work in NLP models	Architecture-level explanation using block diagrams, attention visualization, concept-based discussion	Illustration of self-attention weight computation and comparison with recurrent models.
		Transformers(multi-head attention)	T-1	SW-7 SW-8	This lecture introduces the Transformer architecture, focusing on self-attention and multi head attention mechanisms that enable parallel sequence modeling in NLP.	Students will be able to explain the Transformer architecture and describe how self attention and multi head attention mechanisms work in NLP models	Architecture-level explanation using block diagrams, attention visualization, concept-based discussion	

Week 10	Lecture 20	Transformers(positional encoding)	T-1		This lecture explains positional encoding and the internal structure of transformer encoder and decoder blocks, along with subword tokenization methods such as Byte-Pair Encoding and WordPiece.	Students will be able to explain positional encoding, describe transformer encoder–decoder blocks, and apply subword tokenization methods such as BPE and WordPiece.	Diagram-based explanation, walkthrough of transformer blocks, demonstration of tokenization techniques.	Visualizing positional encodings and demonstrating BPE/WordPiece tokenization on sample text.
		Transformers(transformer encoder and decoder blocks)	T-1		This lecture explains positional encoding and the internal structure of transformer encoder and decoder blocks, along with subword tokenization methods such as Byte-Pair Encoding and WordPiece.	Students will be able to explain positional encoding, describe transformer encoder–decoder blocks, and apply subword tokenization methods such as BPE and WordPiece.	Diagram-based explanation, walkthrough of transformer blocks, demonstration of tokenization techniques.	Visualizing positional encodings and demonstrating BPE/WordPiece tokenization on sample text.
		Transformers(tokenization methods)	T-1	SW-4 SW-5 SW-6	This lecture explains positional encoding and the internal structure of transformer encoder and decoder blocks, along with subword tokenization methods such as Byte-Pair Encoding and WordPiece.	Students will be able to explain positional encoding, describe transformer encoder–decoder blocks, and apply subword tokenization methods such as BPE and WordPiece.	Diagram-based explanation, walkthrough of transformer blocks, demonstration of tokenization techniques.	
		Transformers(Byte-Pair Encoding and WordPiece)	T-1		This lecture explains positional encoding and the internal structure of transformer encoder and decoder blocks, along with subword tokenization methods such as Byte-Pair Encoding and WordPiece.	Students will be able to explain positional encoding, describe transformer encoder–decoder blocks, and apply subword tokenization methods such as BPE and WordPiece.	Diagram-based explanation, walkthrough of transformer blocks, demonstration of tokenization techniques.	Visualizing positional encodings and demonstrating BPE/WordPiece tokenization on sample text.
Week 11	Lecture 21				Project			

Week 11	Lecture 22	Pretrained Language Models (pretrained transformer models)	T-1 R-1			This lecture explains pretraining objectives such as masked language modeling, next sentence prediction, and causal language modeling, and demonstrates transfer learning and fine-tuning of pretrained models for NLP tasks	Students will be able to explain different pretraining objectives and apply transfer learning and fine tuning of pretrained language models for NLP tasks such as text classification, NER, and question answering.	Objective-wise explanation, fine tuning workflow walkthrough, demonstration using HuggingFace Transforme4
		Pretrained Language Models (BERT)	T-1 R-1			This lecture explains pretraining objectives such as masked language modeling, next sentence prediction, and causal language modeling, and demonstrates transfer learning and fine-tuning of pretrained models for NLP tasks	Students will be able to explain different pretraining objectives and apply transfer learning and fine tuning of pretrained language models for NLP tasks such as text classification, NER, and question answering.	Objective-wise explanation, fine tuning workflow walkthrough, demonstration using HuggingFace Transforme4
		Pretrained Language Models (GPT)	T-1 R-1	VL-6 VL-7 VL-8		This lecture explains pretraining objectives such as masked language modeling, next sentence prediction, and causal language modeling, and demonstrates transfer learning and fine-tuning of pretrained models for NLP tasks	Students will be able to explain different pretraining objectives and apply transfer learning and fine tuning of pretrained language models for NLP tasks such as text classification, NER, and question answering.	

Week 11	Lecture 22	Pretrained Language Models (T5)	T-1 R-1			This lecture explains pretraining objectives such as masked language modeling, next sentence prediction, and causal language modeling, and demonstrates transfer learning and fine-tuning of pretrained models for NLP tasks	Students will be able to explain different pretraining objectives and apply transfer learning and fine tuning of pretrained language models for NLP tasks such as text classification, NER, and question answering.	Objective-wise explanation, fine tuning workflow walkthrough, demonstration using HuggingFace Transforme4
		Pretrained Language Models (masked language modeling)	T-1 R-1			This lecture explains pretraining objectives such as masked language modeling, next sentence prediction, and causal language modeling, and demonstrates transfer learning and fine-tuning of pretrained models for NLP tasks	Students will be able to explain different pretraining objectives and apply transfer learning and fine tuning of pretrained language models for NLP tasks such as text classification, NER, and question answering.	Objective-wise explanation, fine tuning workflow walkthrough, demonstration using HuggingFace Transforme4
		Pretrained Language Models (next sentence prediction)	T-1 R-1			This lecture explains pretraining objectives such as masked language modeling, next sentence prediction, and causal language modeling, and demonstrates transfer learning and fine-tuning of pretrained models for NLP tasks	Students will be able to explain different pretraining objectives and apply transfer learning and fine tuning of pretrained language models for NLP tasks such as text classification, NER, and question answering.	Objective-wise explanation, fine tuning workflow walkthrough, demonstration using HuggingFace Transforme4

Week 11	Lecture 22	Pretrained Language Models (causal language modeling)	T-1 R-1			This lecture explains pretraining objectives such as masked language modeling, next sentence prediction, and causal language modeling, and demonstrates transfer learning and fine-tuning of pretrained models for NLP tasks	Students will be able to explain different pretraining objectives and apply transfer learning and fine tuning of pretrained language models for NLP tasks such as text classification, NER, and question answering.	Objective-wise explanation, fine tuning workflow walkthrough, demonstration using HuggingFace Transforme4
		Pretrained Language Models (transfer learning for NLP tasks)	T-1 R-1			This lecture explains pretraining objectives such as masked language modeling, next sentence prediction, and causal language modeling, and demonstrates transfer learning and fine-tuning of pretrained models for NLP tasks	Students will be able to explain different pretraining objectives and apply transfer learning and fine tuning of pretrained language models for NLP tasks such as text classification, NER, and question answering.	Objective-wise explanation, fine tuning workflow walkthrough, demonstration using HuggingFace Transforme4
		Pretrained Language Models (fine-tuning for text classification)	T-1 R-1			This lecture explains pretraining objectives such as masked language modeling, next sentence prediction, and causal language modeling, and demonstrates transfer learning and fine-tuning of pretrained models for NLP tasks	Students will be able to explain different pretraining objectives and apply transfer learning and fine tuning of pretrained language models for NLP tasks such as text classification, NER, and question answering.	Objective-wise explanation, fine tuning workflow walkthrough, demonstration using HuggingFace Transforme4

Week 11	Lecture 22	Pretrained Language Models (named entity recognition)	T-1 R-1			This lecture explains pretraining objectives such as masked language modeling, next sentence prediction, and causal language modeling, and demonstrates transfer learning and fine-tuning of pretrained models for NLP tasks	Students will be able to explain different pretraining objectives and apply transfer learning and fine tuning of pretrained language models for NLP tasks such as text classification, NER, and question answering.	Objective-wise explanation, fine tuning workflow walkthrough, demonstration using HuggingFace Transforme4
		Pretrained Language Models (question answering)	T-1 R-1			This lecture explains pretraining objectives such as masked language modeling, next sentence prediction, and causal language modeling, and demonstrates transfer learning and fine-tuning of pretrained models for NLP tasks	Students will be able to explain different pretraining objectives and apply transfer learning and fine tuning of pretrained language models for NLP tasks such as text classification, NER, and question answering.	Objective-wise explanation, fine tuning workflow walkthrough, demonstration using HuggingFace Transforme4
		Pretrained Language Models (HuggingFace Transformers)	T-1 R-1			This lecture explains pretraining objectives such as masked language modeling, next sentence prediction, and causal language modeling, and demonstrates transfer learning and fine-tuning of pretrained models for NLP tasks	Students will be able to explain different pretraining objectives and apply transfer learning and fine tuning of pretrained language models for NLP tasks such as text classification, NER, and question answering.	Objective-wise explanation, fine tuning workflow walkthrough, demonstration using HuggingFace Transforme4
Week 12	Lecture 23				Test			

Week 12	Lecture 24	Generative NLP and LLMs (generative NLP models)	T-1 R-1		This lecture introduces generative NLP models and large language models, explaining fundamental text generation strategies used in modern NLP systems.	Students will be able to explain generative NLP models and large language models and describe basic strategies used for text generation.	Conceptual explanation using model diagrams, discussion on LLM capabilities, example-based teaching.	Text generation examples using large language models for short text completion.
		Generative NLP and LLMs (text generation strategies)	T-1 R-1		This lecture introduces generative NLP models and large language models, explaining fundamental text generation strategies used in modern NLP systems.	Students will be able to explain generative NLP models and large language models and describe basic strategies used for text generation.	Conceptual explanation using model diagrams, discussion on LLM capabilities, example-based teaching.	Text generation examples using large language models for short text completion.
Week 13	Lecture 25	Generative NLP and LLMs (greedy search)	T-1 R-1		This lecture explains decoding strategies used in text generation, including greedy search, beam search, top-k sampling, and nucleus sampling.	Students will be able to compare different decoding strategies and apply suitable text generation methods for generative language models.	Comparative explanation using decoding trees, numerical examples, and visualization of generation paths.	Comparison of generated text outputs using greedy, beam search, top-k sampling, and nucleus sampling.
		Generative NLP and LLMs (beam search)	T-1 R-1		This lecture explains decoding strategies used in text generation, including greedy search, beam search, top-k sampling, and nucleus sampling.	Students will be able to compare different decoding strategies and apply suitable text generation methods for generative language models.	Comparative explanation using decoding trees, numerical examples, and visualization of generation paths.	Comparison of generated text outputs using greedy, beam search, top-k sampling, and nucleus sampling.
		Generative NLP and LLMs (top-k)	T-1 R-1		This lecture explains decoding strategies used in text generation, including greedy search, beam search, top-k sampling, and nucleus sampling.	Students will be able to compare different decoding strategies and apply suitable text generation methods for generative language models.	Comparative explanation using decoding trees, numerical examples, and visualization of generation paths.	Comparison of generated text outputs using greedy, beam search, top-k sampling, and nucleus sampling.

Week 13	Lecture 25	Generative NLP and LLMs (nucleus sampling)	T-1 R-1		This lecture explains decoding strategies used in text generation, including greedy search, beam search, top-k sampling, and nucleus sampling.	Students will be able to compare different decoding strategies and apply suitable text generation methods for generative language models.	Comparative explanation using decoding trees, numerical examples, and visualization of generation paths.	Comparison of generated text outputs using greedy, beam search, top-k sampling, and nucleus sampling.
	Lecture 26	Generative NLP and LLMs (instruction-tuned)	T-1 R-1		This lecture introduces instruction-tuned large language models and explains their use in dialogue generation and reasoning tasks.	Students will be able to explain instruction-tuned language models and describe how large language models perform dialogue generation and reasoning tasks.	Conceptual explanation with real-world examples, discussion on LLM behavior, demonstration based learning.	Dialogue generation and reasoning examples using instruction tuned large language models.
		Generative NLP and LLMs (large language models)	T-1 R-1		This lecture introduces instruction-tuned large language models and explains their use in dialogue generation and reasoning tasks.	Students will be able to explain instruction-tuned language models and describe how large language models perform dialogue generation and reasoning tasks.	Conceptual explanation with real-world examples, discussion on LLM behavior, demonstration based learning.	Dialogue generation and reasoning examples using instruction tuned large language models.
		Generative NLP and LLMs (dialogue generation)	T-1 R-1	SW-9 SW-10 SW-11	This lecture introduces instruction-tuned large language models and explains their use in dialogue generation and reasoning tasks.	Students will be able to explain instruction-tuned language models and describe how large language models perform dialogue generation and reasoning tasks.	Conceptual explanation with real-world examples, discussion on LLM behavior, demonstration based learning.	
		Generative NLP and LLMs (reasoning tasks)	T-1 R-1		This lecture introduces instruction-tuned large language models and explains their use in dialogue generation and reasoning tasks.	Students will be able to explain instruction-tuned language models and describe how large language models perform dialogue generation and reasoning tasks.	Conceptual explanation with real-world examples, discussion on LLM behavior, demonstration based learning.	Dialogue generation and reasoning examples using instruction tuned large language models.

Week 14	Lecture 27	Generative NLP and LLMs (model behaviors in summarization)	T-1 R-1	SW-12 SW-13 SW-14	This lecture discusses the behavior of large language models in summarization tasks and explains evaluation metrics, perplexity, human judgment measures, and challenges such as explainability and hallucination in LLMs.	Students will be able to evaluate generative language models, analyze model behavior in summarization, and identify explainability and hallucination challenges in large language models	Discussion-based learning, metric driven evaluation examples, case study analysis on hallucination in LLMs	
		Generative NLP and LLMs (evaluation metrics)	T-1 R-1	SW-15 VL-1 VL-2	This lecture discusses the behavior of large language models in summarization tasks and explains evaluation metrics, perplexity, human judgment measures, and challenges such as explainability and hallucination in LLMs.	Students will be able to evaluate generative language models, analyze model behavior in summarization, and identify explainability and hallucination challenges in large language models	Discussion-based learning, metric driven evaluation examples, case study analysis on hallucination in LLMs	
		Generative NLP and LLMs (perplexity and human judgment measures)	T-1 R-1		This lecture discusses the behavior of large language models in summarization tasks and explains evaluation metrics, perplexity, human judgment measures, and challenges such as explainability and hallucination in LLMs.	Students will be able to evaluate generative language models, analyze model behavior in summarization, and identify explainability and hallucination challenges in large language models	Discussion-based learning, metric driven evaluation examples, case study analysis on hallucination in LLMs	Evaluation of generated summaries, comparison of perplexity scores, and examples of hallucinated responses from language models.
		Generative NLP and LLMs (explainability and hallucination in LLMs)	T-1 R-1		This lecture discusses the behavior of large language models in summarization tasks and explains evaluation metrics, perplexity, human judgment measures, and challenges such as explainability and hallucination in LLMs.	Students will be able to evaluate generative language models, analyze model behavior in summarization, and identify explainability and hallucination challenges in large language models	Discussion-based learning, metric driven evaluation examples, case study analysis on hallucination in LLMs	Evaluation of generated summaries, comparison of perplexity scores, and examples of hallucinated responses from language models.

		SPILL OVER						
Week 14	Lecture 28				Spill Over			
Week 15	Lecture 29				Spill Over			
	Lecture 30				Spill Over			

Scheme for CA:

CA Category of this Course Code is:C010102 (Total 3 tasks, 1 compulsory and out of remaining 1 best out of 2 to be considered)

Component	Iscompulsory	Weightage (%)	Mapped CO(s)
Test	NO	33.34	CO1, CO2, CO3, CO4
Test - Code based	NO	33.34	CO1, CO2
Project	Yes	66.66	CO1, CO2, CO3, CO4, CO5, CO6

Details of Academic Task(s)

Academic Task	Objective	Detail of Academic Task	Nature of Academic Task (group/individuals)	Academic Task Mode	Marks	Allottment / submission Week
Project	The objective of this project is to apply the concepts and techniques covered throughout the Deep Learning for Natural Language Processing syllabus to address a real world NLP problem.	Students are required to carry out a comprehensive NLP project covering the entire course syllabus, involving problem identification, dataset selection, text preprocessing, model design and implementation using deep learning and transformer-based approaches, performance evaluation using suitable metrics, and analysis of results, followed by proper documentation and presentation of the work. Evaluation Components : 1. Technical Execution & Skill Demonstration – 10 Marks 2. Innovation & Project Report – 10 Marks 3. Project Presentation & Viva (Q&A) – 10 Marks	Group	Online	30	3 / 12
Test	To test the subject knowledge of the students and provide opportunity to improve their CA performance in case of low marks or any missed CA	Subjective test covering syllabus from unit1 to unit4	Individual	Offline	30	11 / 12

An instruction plan is only a tentative plan. The teacher may make some changes in his/her teaching plan. The students are advised to use syllabus for preparation of all examinations. The students are expected to keep themselves updated on the contemporary issues related to the course. Upto 20% of the questions in any examination/Academic tasks can be asked from such issues even if not explicitly mentioned in the instruction plan.

Test - Code based	The objective of this assessment is to evaluate students' understanding of fundamental concepts in Natural Language Processing covered in Units I–II.	This academic task is an objective-type individual assessment covering Units I–II of Deep Learning for NLP. It includes MCQs, numerical problems, and basic code-based questions on NLP foundations, word embeddings, and deep learning models. The task evaluates students' conceptual clarity, analytical skills, and application of core NLP techniques.	Individual	Offline	30	4 / 5
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MOOCs/ Certification etc. mapped with the Academic Task(s)

Academic Task	Name Of Certification/Online Course/Test/Competition mapped	Type	Offered By Organisation
Test - Code based	DEEP LEARNING FOR NATURAL LANGUAGE PROCESSING, IIT KHARAGPUR BY PROF. PAWAN GOYAL	MOOCs	NPTEL

Where MOOCs/ Certification etc. are mapped with Academic Tasks:

1. Students have choice to appear for Academic Task or MOOCs etc.
2. The student may appear for both, In this case best obtained marks will be considered.

List of suggested topics for term paper[at least 15] (Student to spend about 15 hrs on any one specified term paper)

Sr. No.	Topic
1	Sentiment analysis, text classification, named entity recognition, machine translation, text summarization, chatbot development, question answering, fake news detection, topic modeling, word embedding comparison, transformer-based NLP models, pretrained language model fine-tuning, attention mechanisms, evaluation of NLP models, and bias and explainability in large language models.

Detailed Plan For Practicals

Practical No	Broad topic	Subtopic	Other Readings	Learning Outcomes
Practical 1	List of Practicals	Given a set of text documents, implement a preprocessing pipeline that performs tokenization, lowercasing, noise removal, stopword removal, stemming/lemmatization, and converts the processed text into numerical representations using Bag-of-Words and TF-IDF	OR-1 OR-2 OR-3	Students will be able to implement a complete text preprocessing pipeline and convert raw text documents into numerical feature representations using Bag-of-Words and TF-IDF techniques.

An instruction plan is only a tentative plan. The teacher may make some changes in his/her teaching plan. The students are advised to use syllabus for preparation of all examinations. The students are expected to keep themselves updated on the contemporary issues related to the course. Upto 20% of the questions in any examination/Academic tasks can be asked from such issues even if not explicitly mentioned in the instruction plan.

Practical 2	List of Practicals	Load pretrained Word2Vec or GloVe embeddings, select a set of words, compute their embeddings, and visualize relationships in 2-D space using PCA or t-SNE	AV-4 AV-5 AV-6 VL-4 VL-5 VL-6 VL-7 VL-8	Students will be able to load pretrained word embeddings, analyze semantic relationships between words, and visualize high-dimensional embeddings using PCA or t-SNE.
Practical 3	List of Practicals	Train CBOW and Skip-Gram Word2Vec models on a sample text corpus, compute word similarities and analogies, and compare the performance of both models	RW-1 RW-2 RW-3	Students will be able to train CBOW and Skip-Gram Word2Vec models, compute word similarities and analogies, and compare the performance of both embedding approaches.
Practical 4	List of Practicals	Load pretrained GloVe or Word2Vec embeddings, select word groups belonging to different semantic categories, reduce dimensionality using PCA or t-SNE, and visualize the embeddings to interpret clusters and relationships	OR-4 OR-5 RW-1	Students will be able to analyze semantic word relationships by visualizing pretrained word embeddings using dimensionality reduction techniques such as PCA or t-SNE.
Practical 5	List of Practicals	Build a sentiment analysis model using an LSTM network on a dataset such as the IMDB movie reviews or SMS Spam dataset. Compare its performance with a baseline model that uses Bag-of-Words or TF-IDF features	RW-5 RW-6 RW-7 RW-8	Students will be able to build and evaluate an LSTM-based sentiment analysis model and compare its performance with traditional Bag-of-Words or TF-IDF-based classifiers.
Practical 6	List of Practicals	Implement a Bidirectional LSTM model for Named Entity Recognition using the CoNLL-2003 dataset or a smaller annotated dataset. Train the model to label words as Person, Location, Organization, or Other	SW-1 SW-2 SW-3 SW-4 SW-5 SW-6	Students will be able to implement a Bidirectional LSTM model for Named Entity Recognition and train the model to identify entities such as person, location, and organization.
Practical 7	List of Practicals	Implement a sequence-to-sequence model with Bahdanau or Luong attention to translate short English sentences into another language using a small parallel dataset. Compare translation quality with and without attention	RW-4 RW-5 RW-6	Students will be able to implement a sequence-to-sequence translation model with attention and compare translation quality with and without attention mechanisms.

Practical 8	List of Practicals	Build an attention-based sequence-to-sequence model to generate short summaries from news headlines or paragraph datasets. Train briefly using a subset of the dataset and evaluate generated summaries using ROUGE scores	SW-7 SW-8 SW-9 SW-10 SW-11 SW-12 SW-13 SW-14	Students will be able to build an attention-based sequence-to-sequence model for text summarization and evaluate generated summaries using ROUGE metrics.
Practical 9	List of Practicals	Fine-tune a pretrained BERT model for sentiment classification on a dataset such as the IMDB movie reviews or the SMS Spam dataset. Evaluate the model using accuracy and F1-score, and analyze how BERT's contextual embeddings improve classification performance	RW-5 RW-6 RW-7 RW-8	Students will be able to fine-tune a pretrained BERT model for sentiment classification and analyze how contextual embeddings improve classification performance.
Practical 10	List of Practicals	Use a pretrained GPT-2 model to generate text based on a given prompt. Apply different multiple decoding strategies and compare the fluency, creativity, and coherence of the generated outputs	RW-9 AV-1 AV-2 AV-3	Students will be able to generate text using a pretrained GPT-2 model, apply multiple decoding strategies, and compare the generated outputs in terms of fluency, creativity, and coherence.
	SPILL OVER			
Practical 11	Spill Over			